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Started	Tuesday, 20 October 2026, 10:41 AM
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Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. The parts are merged without sorting
- b. Each part is sorted independently
- c. Each part is searched for a specific element
- d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **2**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort does not guarantee sorted order
- b. Bubble Sort cannot handle duplicate elements
- c. Bubble Sort is less efficient for large lists
- d. Bubble Sort is difficult to implement

The correct answer is: Bubble Sort is less efficient for large lists

Question **3**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort

b. Heap sort

c. Bubble sort

d. Insertion sort

The correct answer is:
Insertion sort

Question **4**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,70,80,90,120,150,200,300,400,470

b. 80,150,200,470,40,90,300,400,70,120

c. 40,80,90,150,200,300,400,470,70,120

d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **5**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Sorting the entire list sequentially
- c. Comparing each element with the others
- d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **6**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Dynamic programming
- b. Greedy method
- c. Divide and conquer
- d. Backtracking

The correct answer is:
Divide and conquer

Question **7**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Rearranging data without sorting
- b. Sorting data in a single pass
- c. Dividing the input into parts, solving each part, and combining the solutions
- d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Merge sort
- b. Selection sort
- c. Heap sort
- d. Quick sort

The correct answer is:
Merge sort

Question **9**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Searching for data in a list
- b. Inserting data into a list
- c. Arranging data in a particular format
- d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting helps in finding duplicates quickly
- b. Sorting decreases the efficiency of selection operations
- c. Sorting makes it harder to search for items
- d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **11**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It bubbles up the largest element in each pass
- b. It divides the list into sublists
- c. It uses the heap data structure
- d. It recursively sorts subproblems

The correct answer is: It bubbles up the largest element in each pass

Question **12**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Rearranging
- b. Merging
- c. Searching
- d. Sorting

The correct answer is: Sorting

Question **13**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sort()
- b. sorted()
- c. arrange()
- d. order()

The correct answer is: sorted()

Question **14**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are split again into smaller sublists
- b. They are compared element by element
- c. They are combined to form a single sorted list
- d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **15**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Sorting
- b. Complexity
- c. Merging
- d. Searching

The correct answer is:
Complexity